

STM - Exercise 1

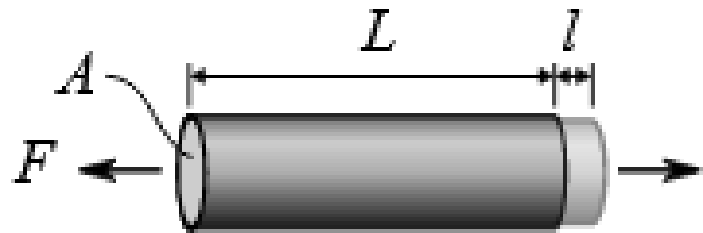
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Exercise 1

Use figure 1 to:

- a) define which test and which class of materials is referred to and why;
- b) calculate elastic modulus,
- c) Calculate the yield strength (0.2 % offset) and the tensile strength,
- d) Calculate the elongation at fracture for a 50 mm long specimen (write on curves data you use).
- e) Calculate the length of the specimen when you keep it loaded in A.



$$\sigma = \frac{F}{A}$$

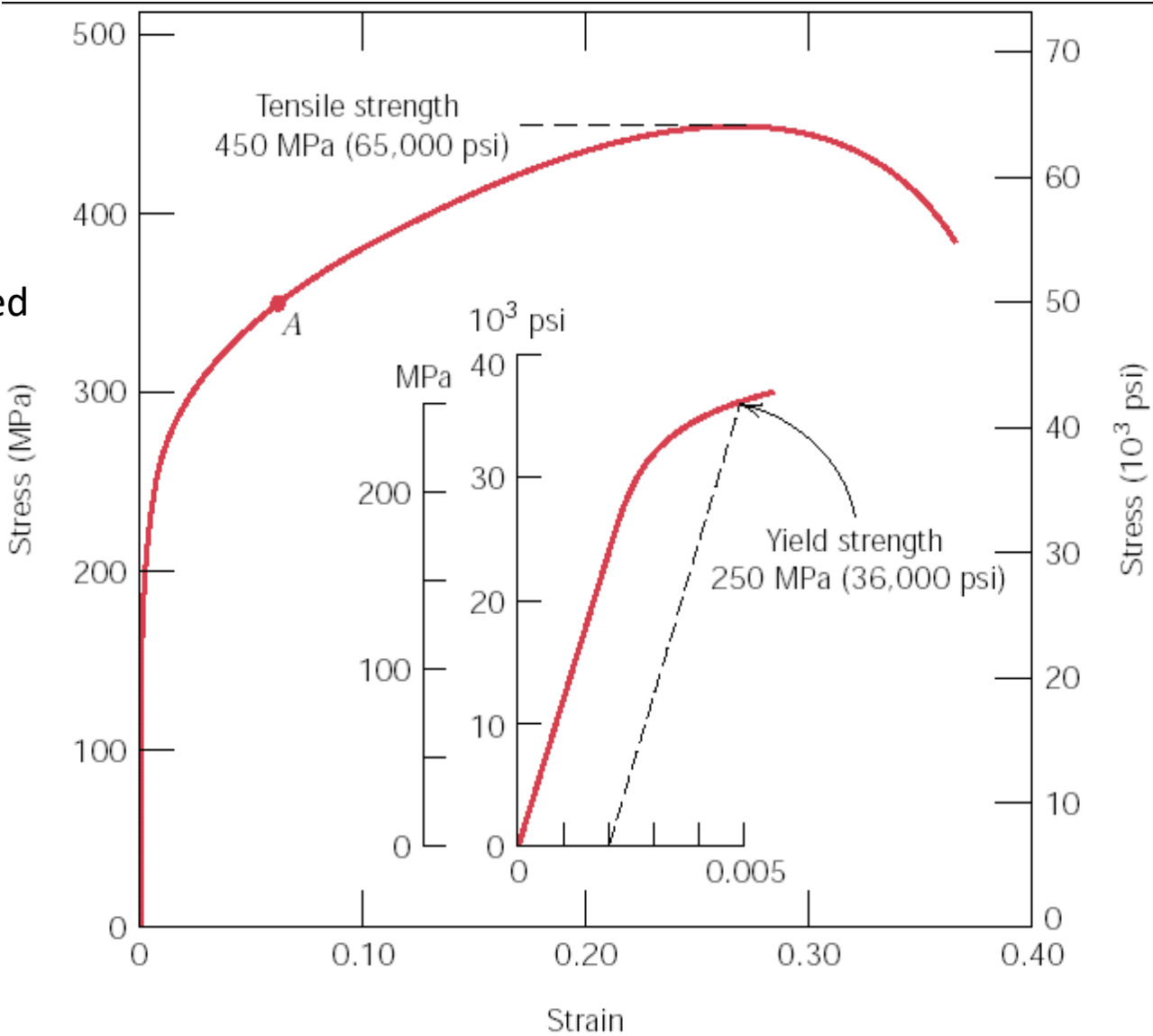
where

- σ stress [Pa]
- F applied force [N]
- A cross-sectional area [m²]

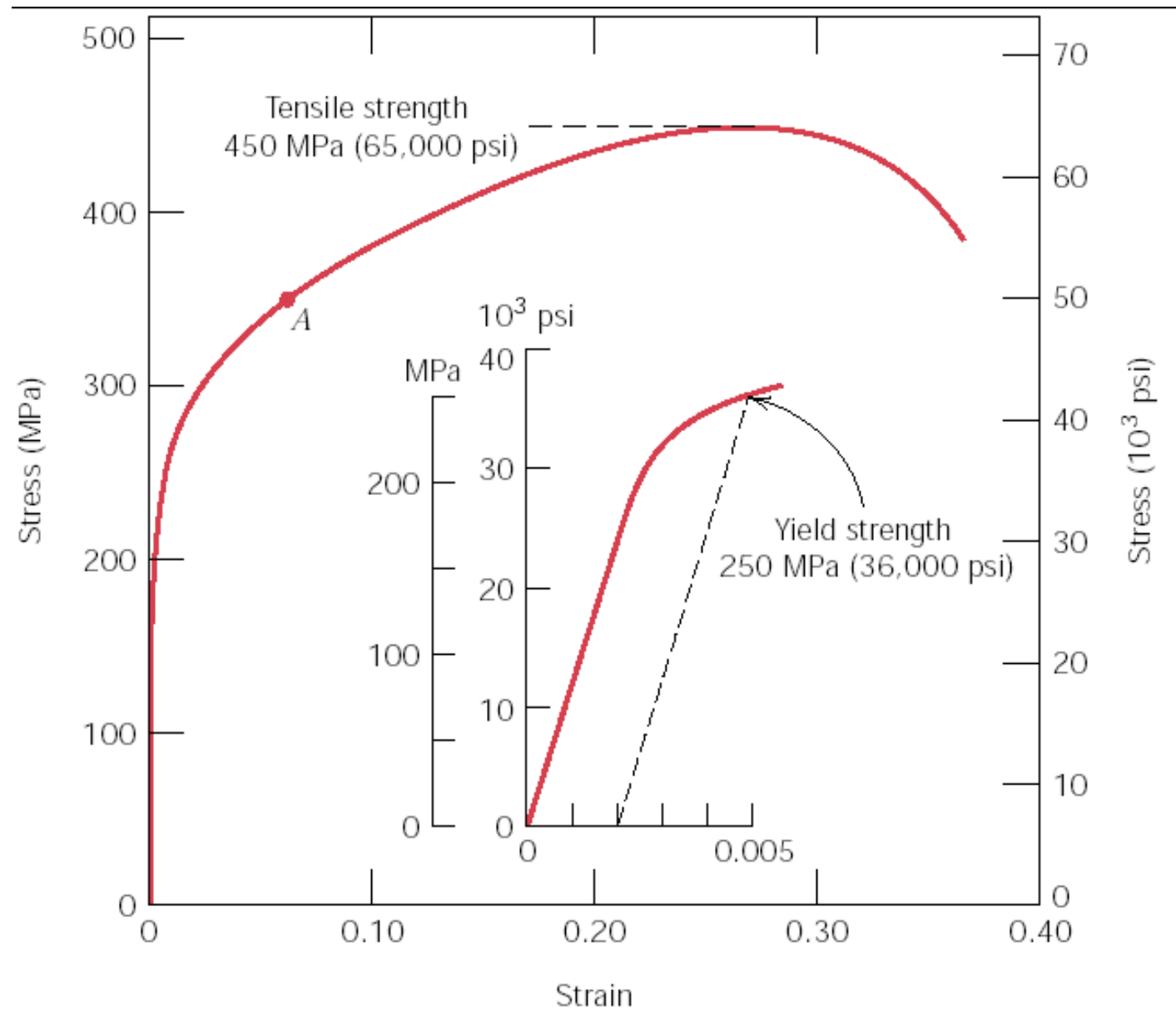
$$\varepsilon = \frac{\Delta L}{L_0}$$

where

- ε strain
- ΔL total elongation [m]
- L_0 original length [m]

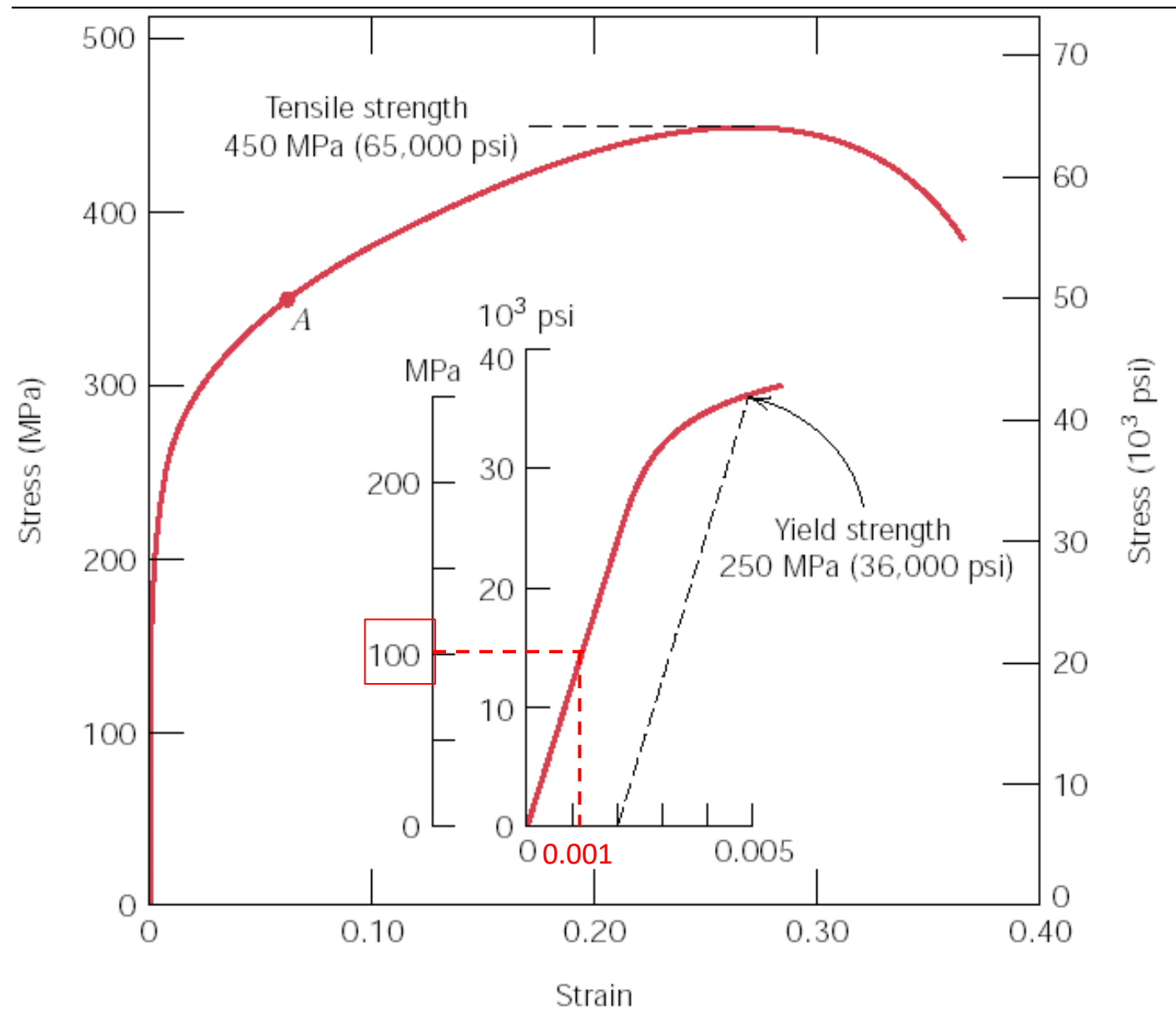


Exercise 1a: Use figure 1 to define which test and which class of materials is referred to and why



Typical stress-strain curve from tensile test
Ductile behavior and stress-strain values typical of
metallic materials

Exercise 1b: Calculate elastic modulus

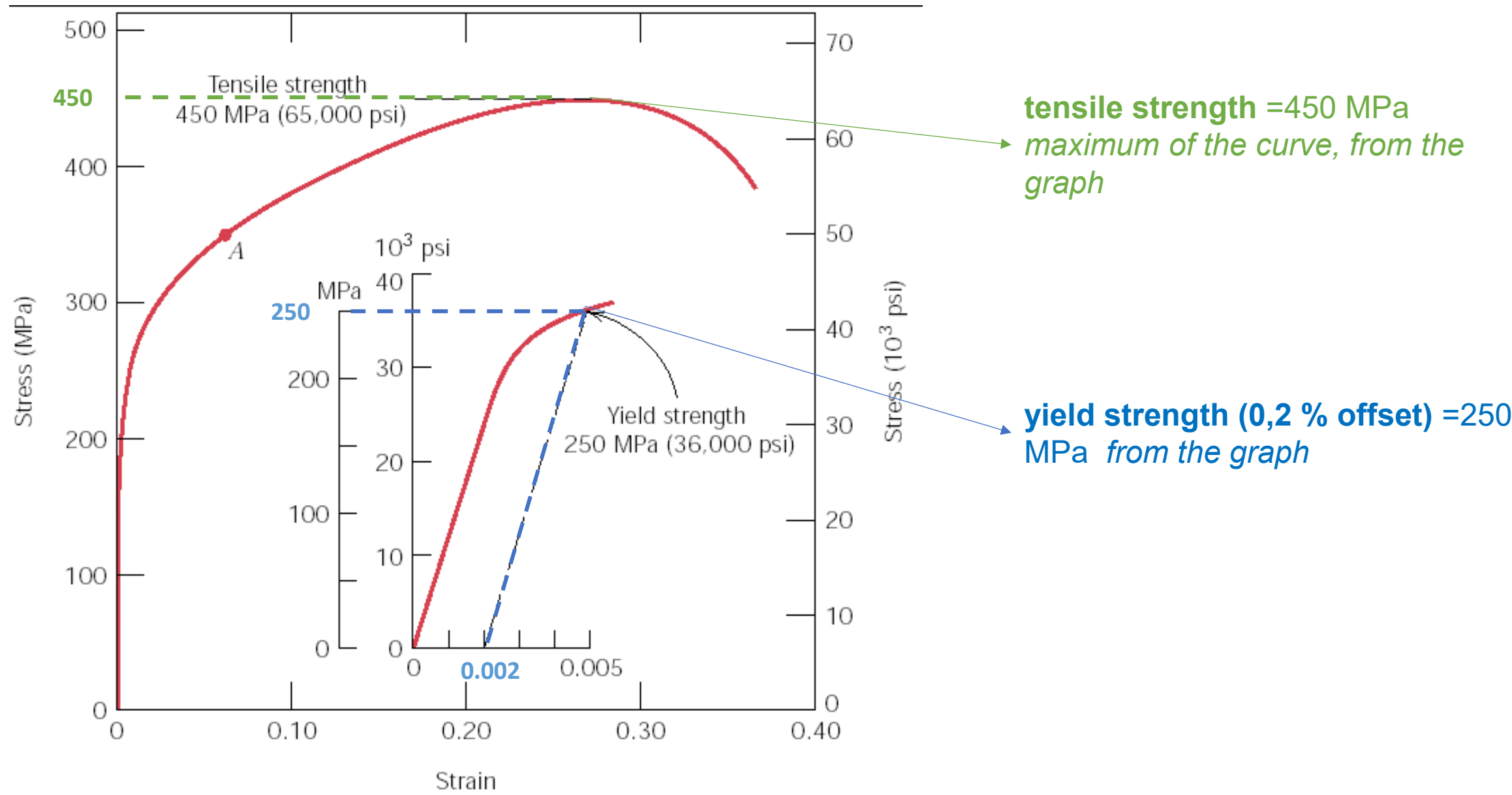


$$E = \frac{\sigma}{\epsilon}$$

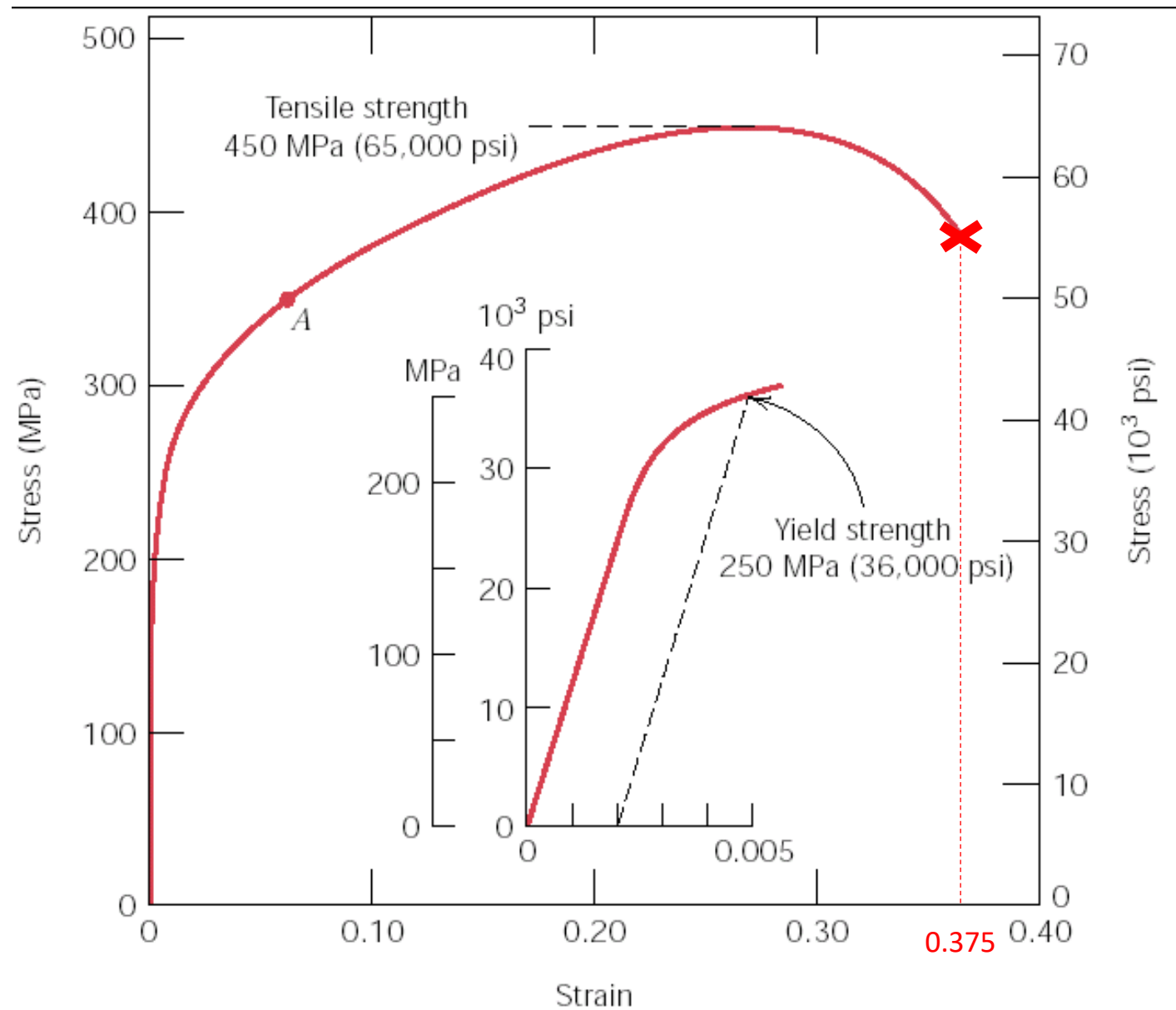
From the linear region we can calculate:

$$E = 100\text{MPa}/0.001 = 100000\text{ Mpa} = 100\text{ Gpa}$$

Exercise 1c: calculate yield strength (0.2 % offset) and tensile strength



Exercise 1d: Calculate elongation at fracture for a 50 mm long specimen (write on curves data you use).

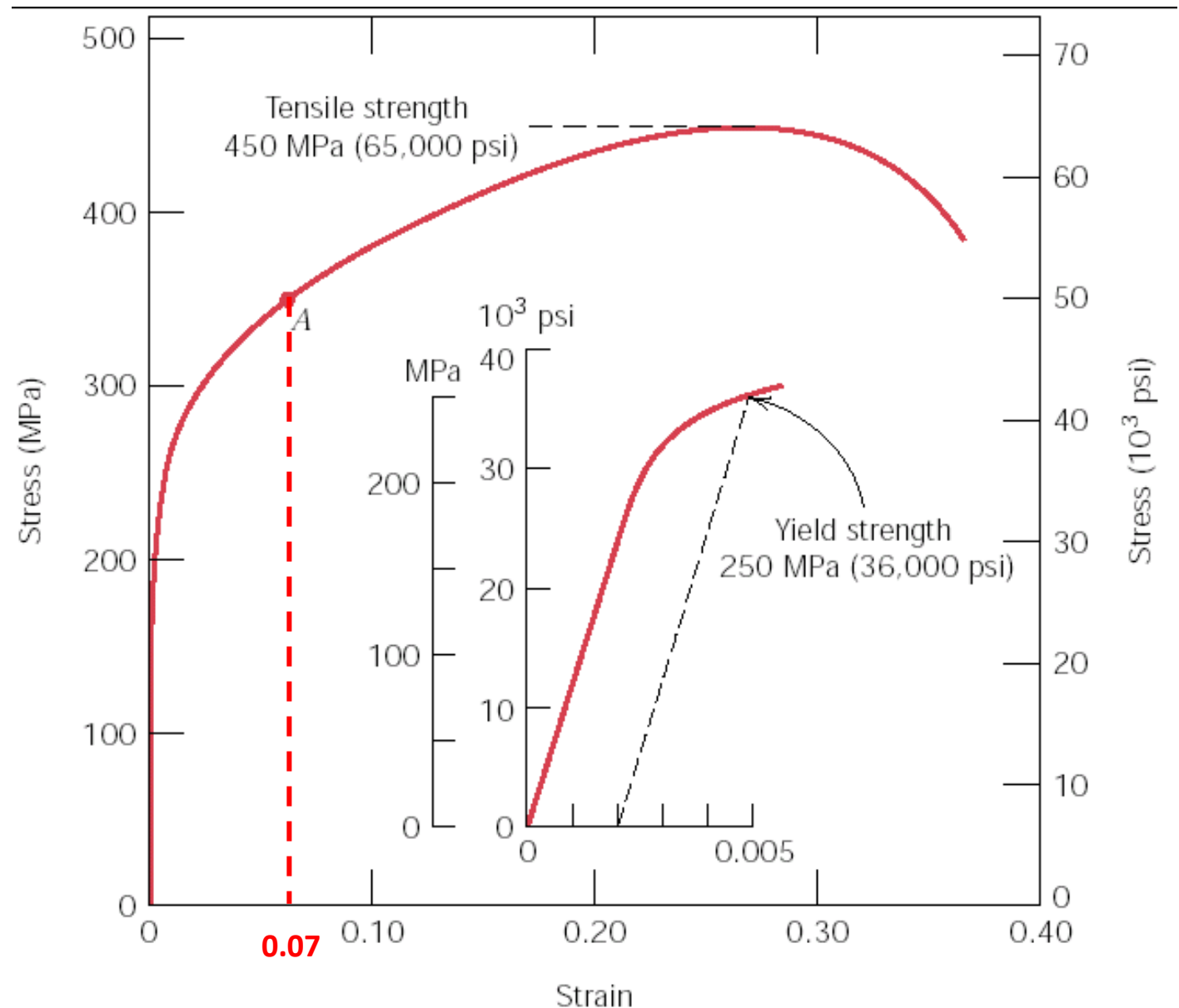


$$\epsilon_F = 0.375$$

$$\epsilon_F = \frac{\Delta l_F}{l_0}$$

$$\Delta l_F = \epsilon_F * l_0 = 0.375 * 50 \text{ mm} \\ = 18.75 \text{ mm}$$

Exercise 1e- Calculate the length of the specimen when you keep it loaded in A.



If you keep the sample loaded
there's no recovery of elastic
deformation

$$\epsilon_A = 0.07$$

$$\epsilon_A = \frac{\Delta l_A}{l_0}$$

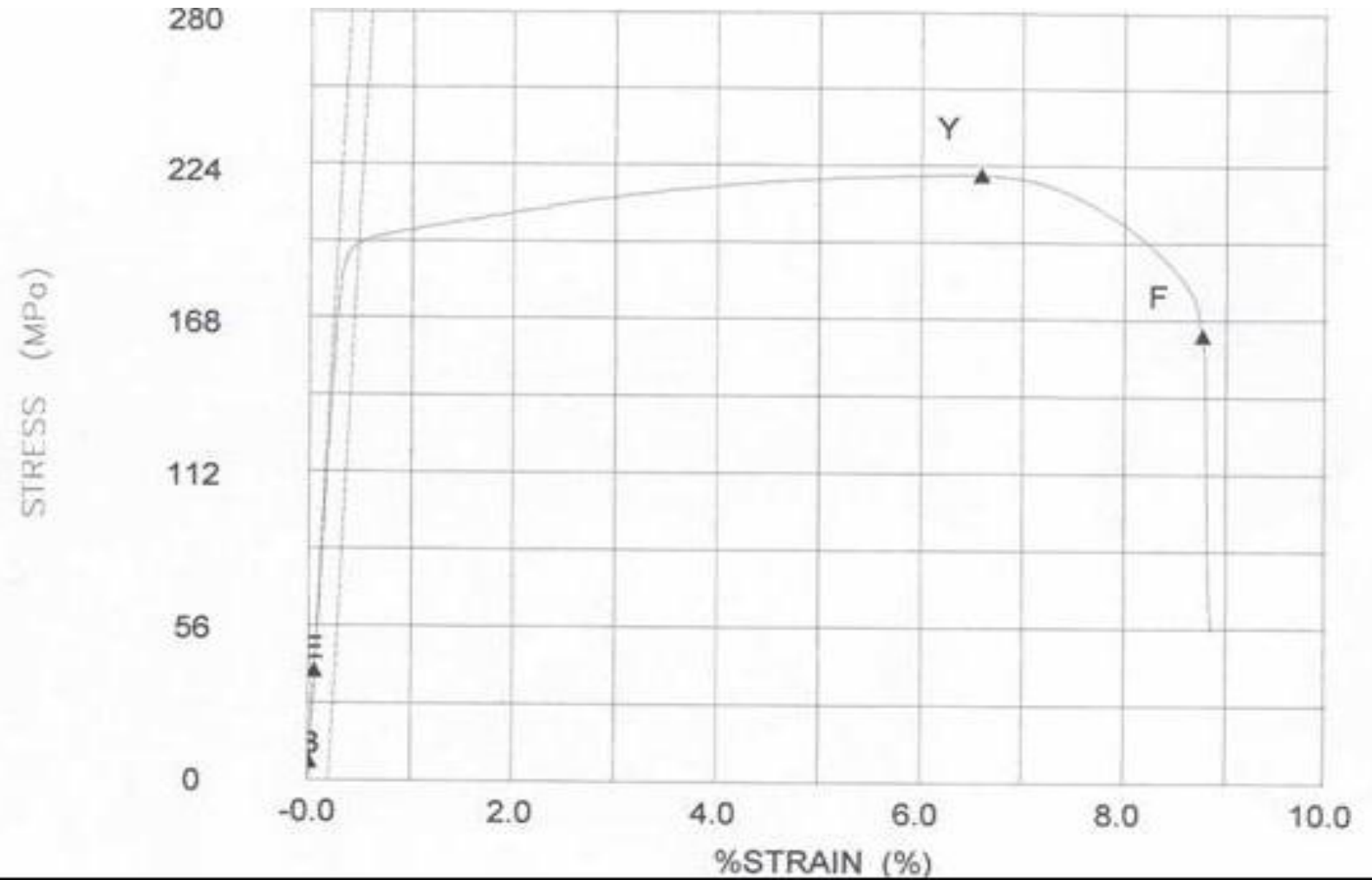
$$\Delta l_A = \epsilon_A * l_0 = 0.07 * 50 \text{ mm} \\ = 3.5 \text{ mm}$$

$$l_A = \Delta l_A + l_0 = 50 \text{ mm} + 3.5 \text{ mm} \\ = 53.5 \text{ mm}$$

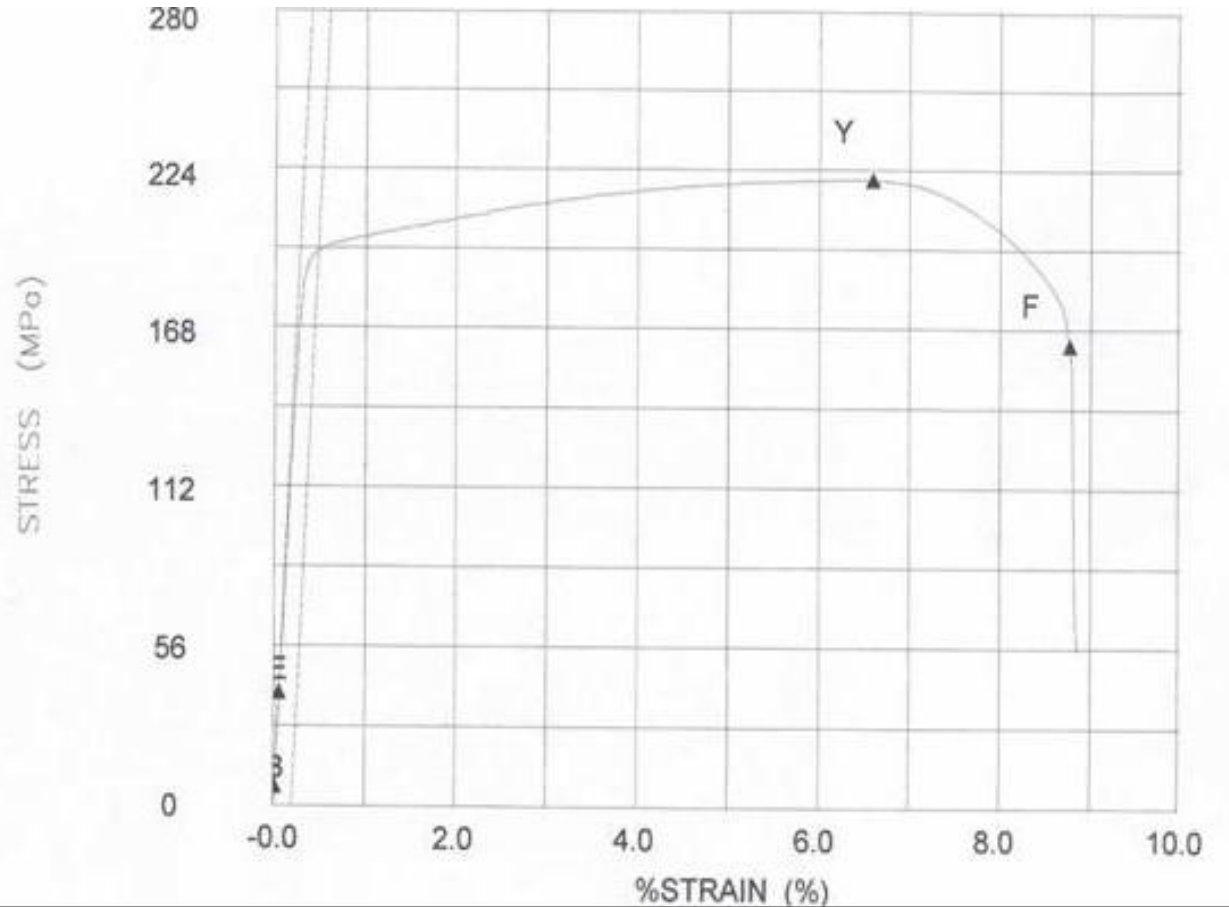
Exercise 2: Use the graph below to define

- which test and which class of materials is referred to and why;
- to calculate elastic modulus,
- To calculate the yield strength (0,2 % strain) and the tensile strength,
- To calculate the elongation at fracture for a 50 mm long specimen;
- To calculate the elongation of this sample when loaded at 200 MPa, then unloaded.

(write on graph stress and strain
you use to calculate).

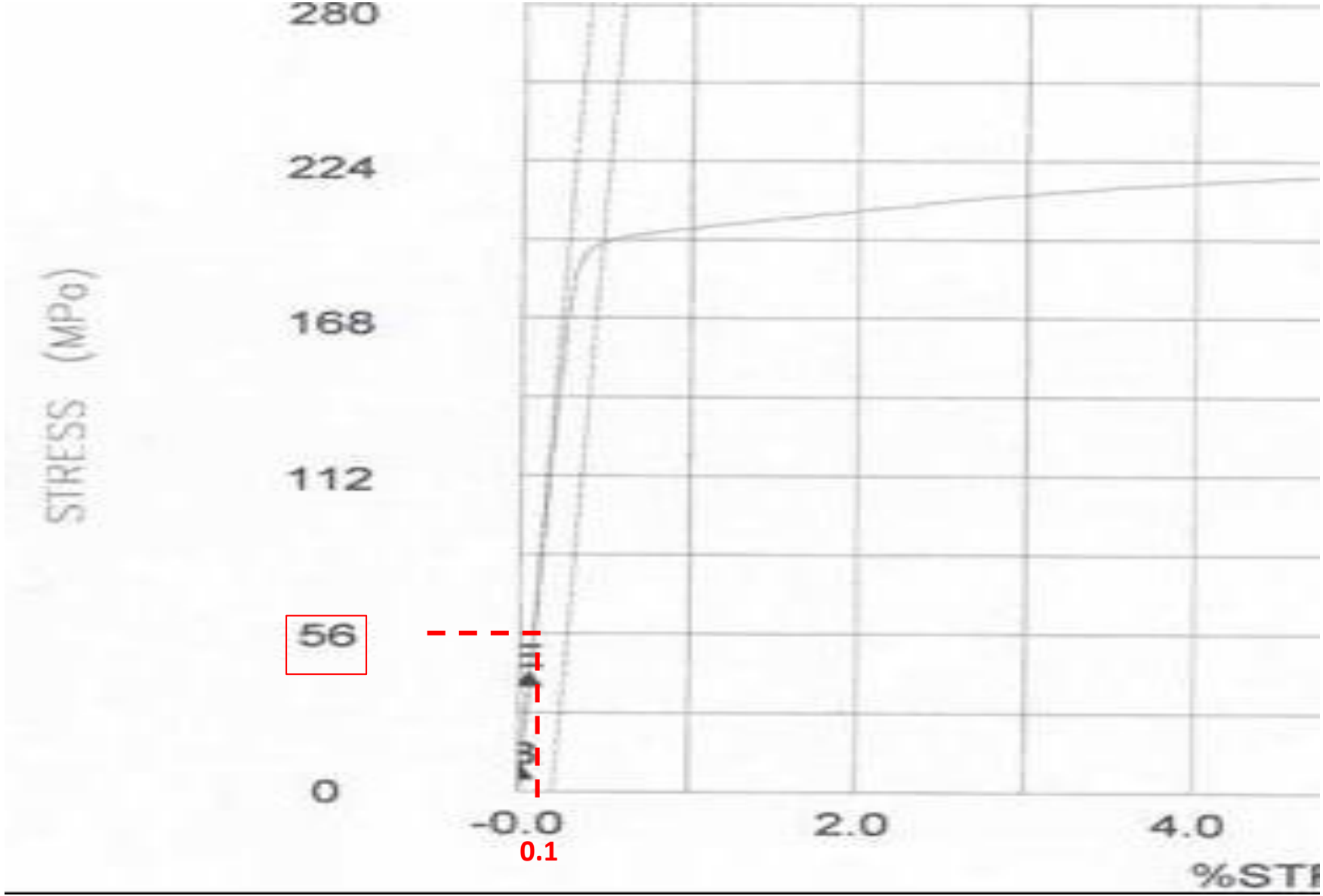


Exercise 2a: Define which test and which class of materials is referred to and why



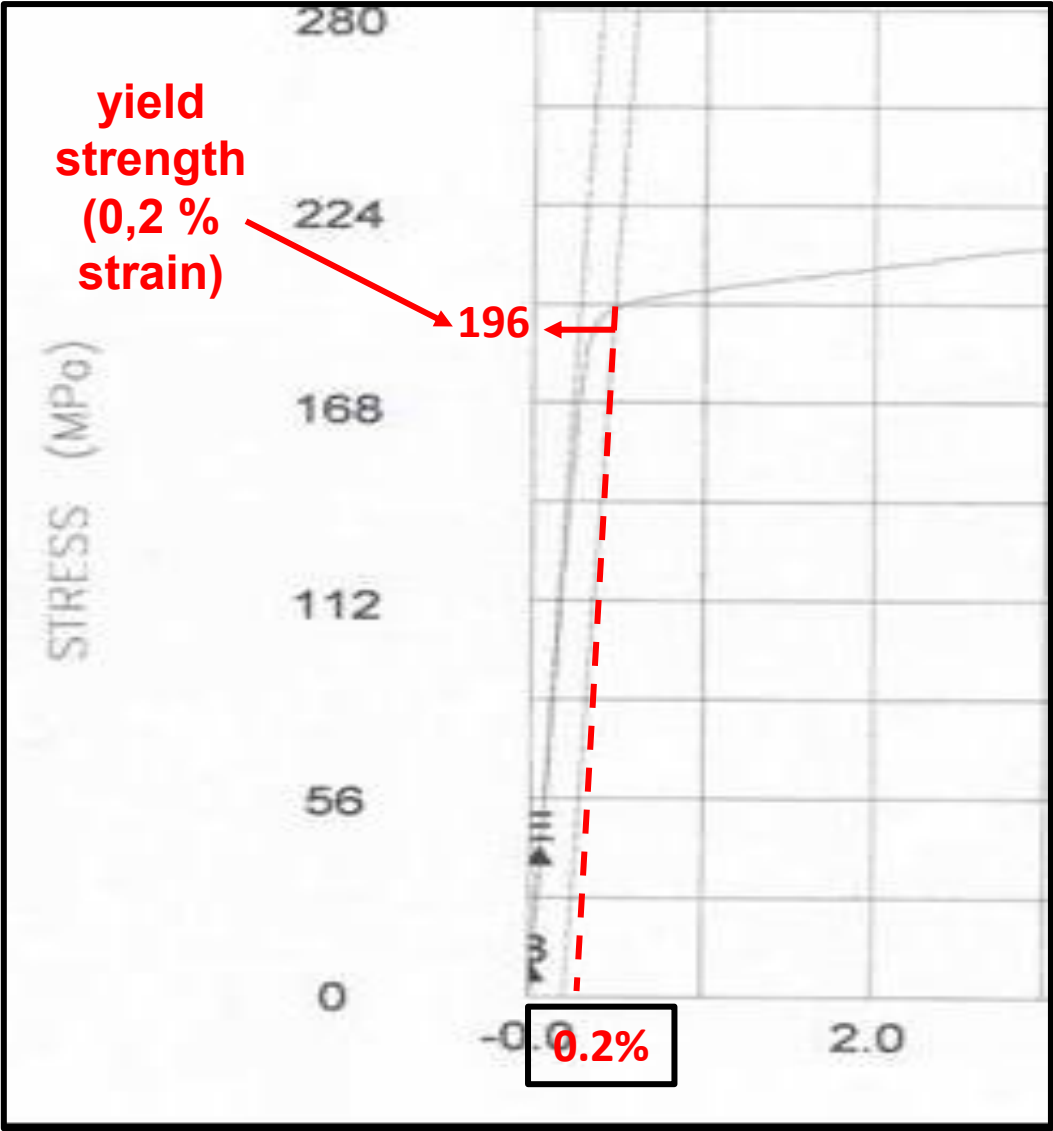
Typical stress-strain curve from tensile test
Ductile behavior and stress-strain values typical
of metallic materials

Exercise 2b: calculate elastic modulus



$\epsilon=0.1\%=0.001$
 $E=\sigma/\epsilon=$
 $56/0.001=56000\text{MPa}$
 $=56\text{GPa}$

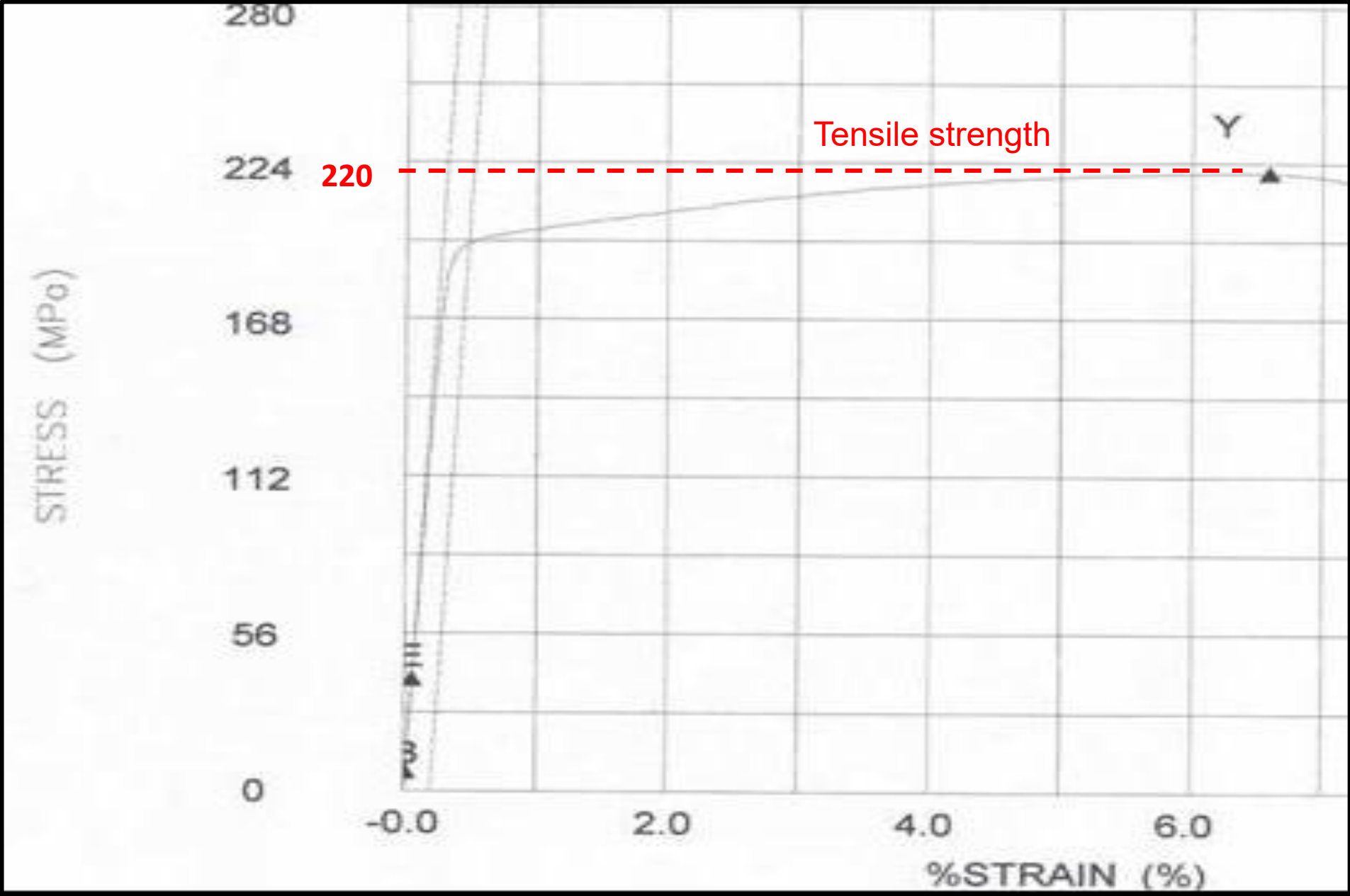
Exercise 2c: calculate yield strength (0,2 % strain) and tensile strength



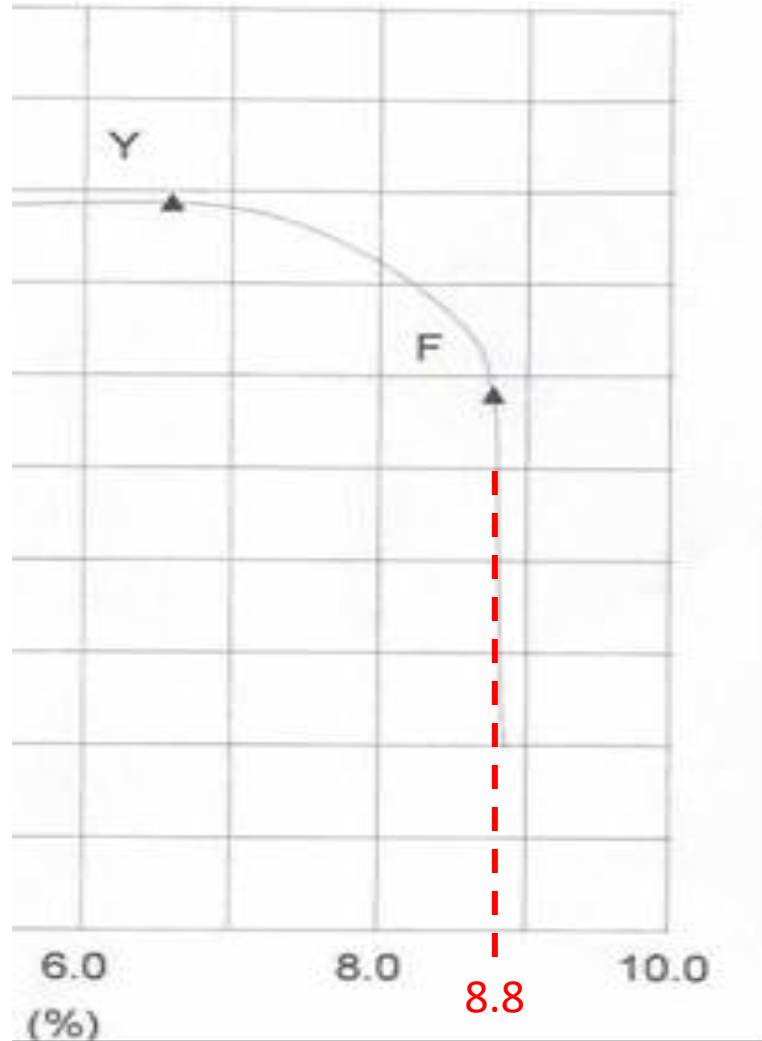
yield strength (0,2 % strain) =196 MPa from the graph

Exercise 2c: calculate yield strength (0,2 % strain) and tensile strength

tensile
strength
(maximum)
=220 MPa
from the graph



Exercise 2d: Calculate elongation at fracture for a 50 mm long specimen

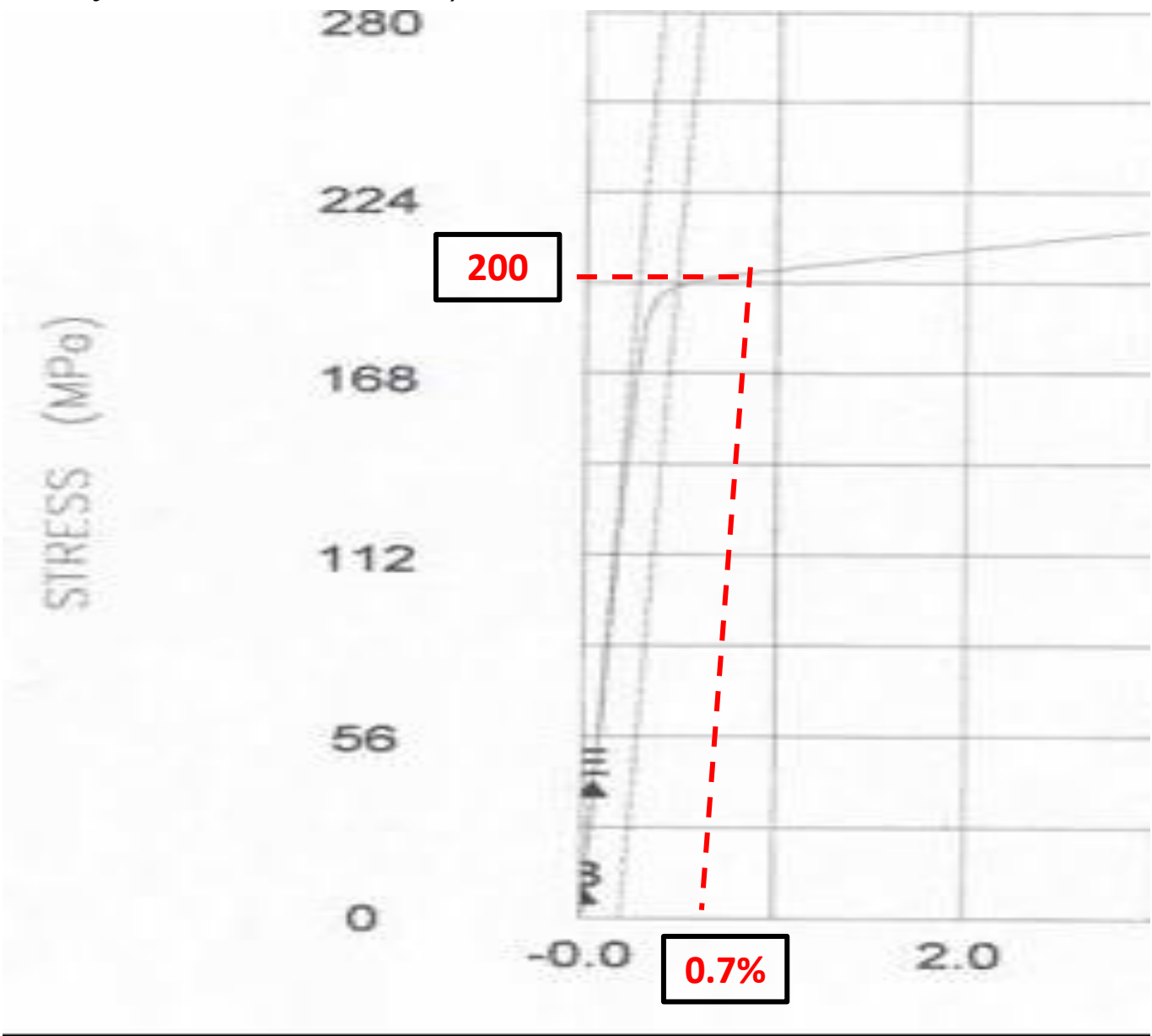


$$\epsilon_F = 8.8\% = 0.088$$

$$\epsilon_F = \Delta l_F / l_0$$

$$\Delta l_F = \epsilon_F * l_0 = 0.088 * 50 = 4.4 \text{ mm}$$

Exercise 2e: Calculate the elongation of this sample when loaded at 200 MPa, then unloaded. (write on graph stress and strain you use to calculate).

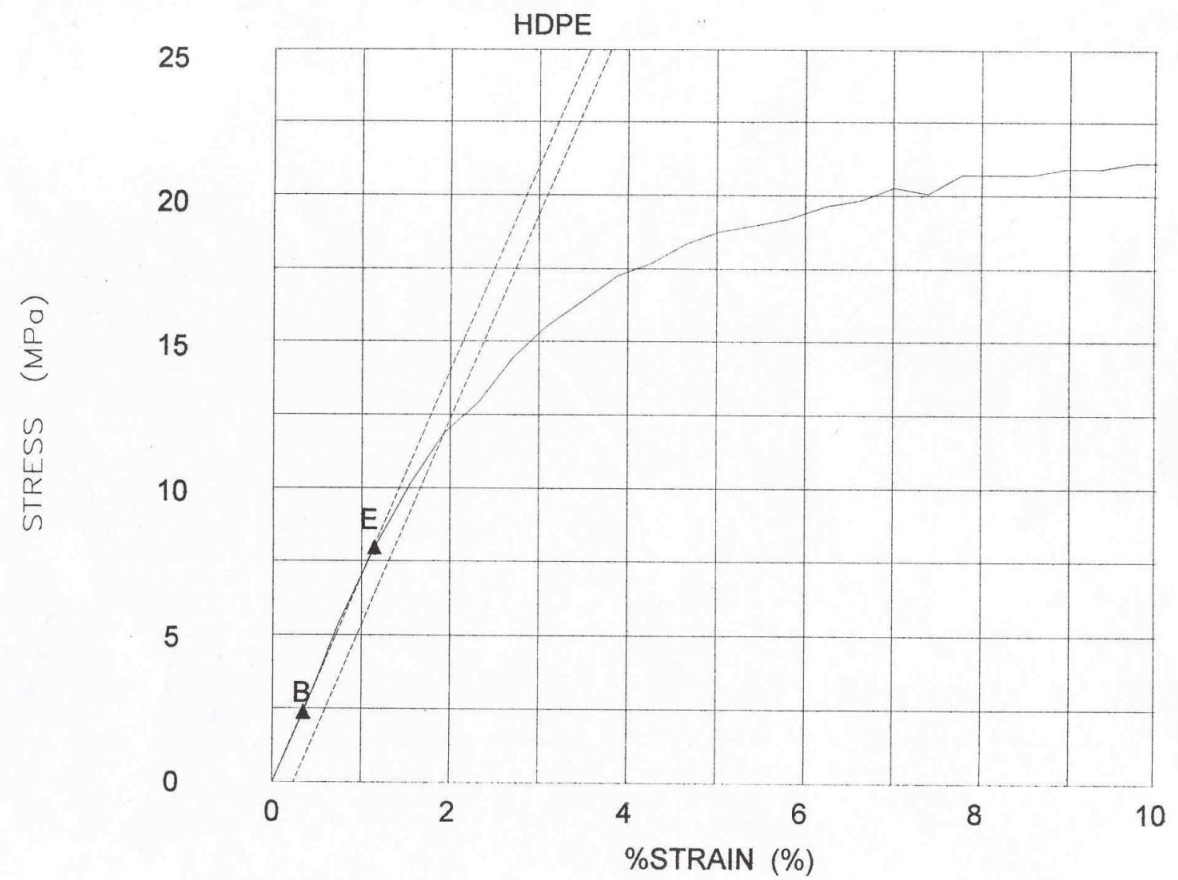
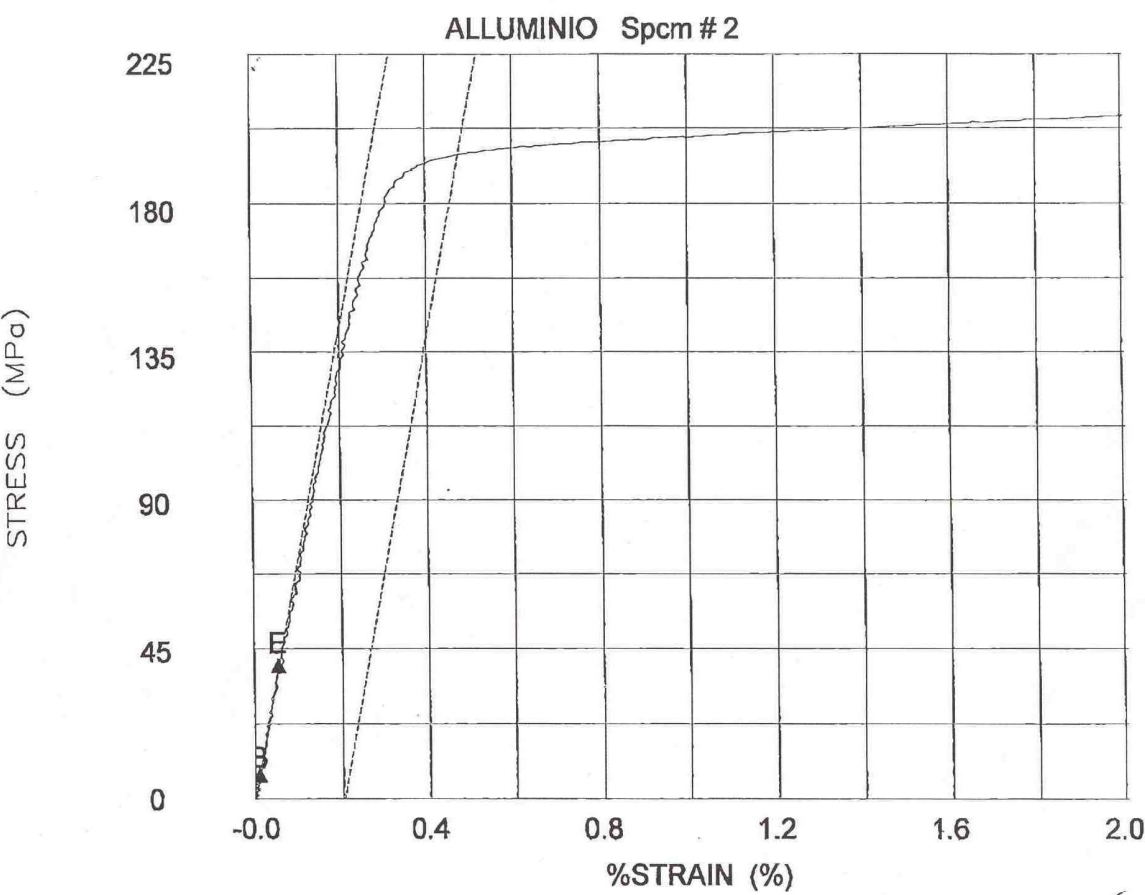


At the unloading after the loading at 200MPa we have → recovery of elastic deformation, and a permanent deformation:

$$\epsilon_p = 0.7\% = 0.007$$
$$\Delta l = \epsilon_p * l_0 = 0.007 * 50 = 0.35 \text{ mm}$$

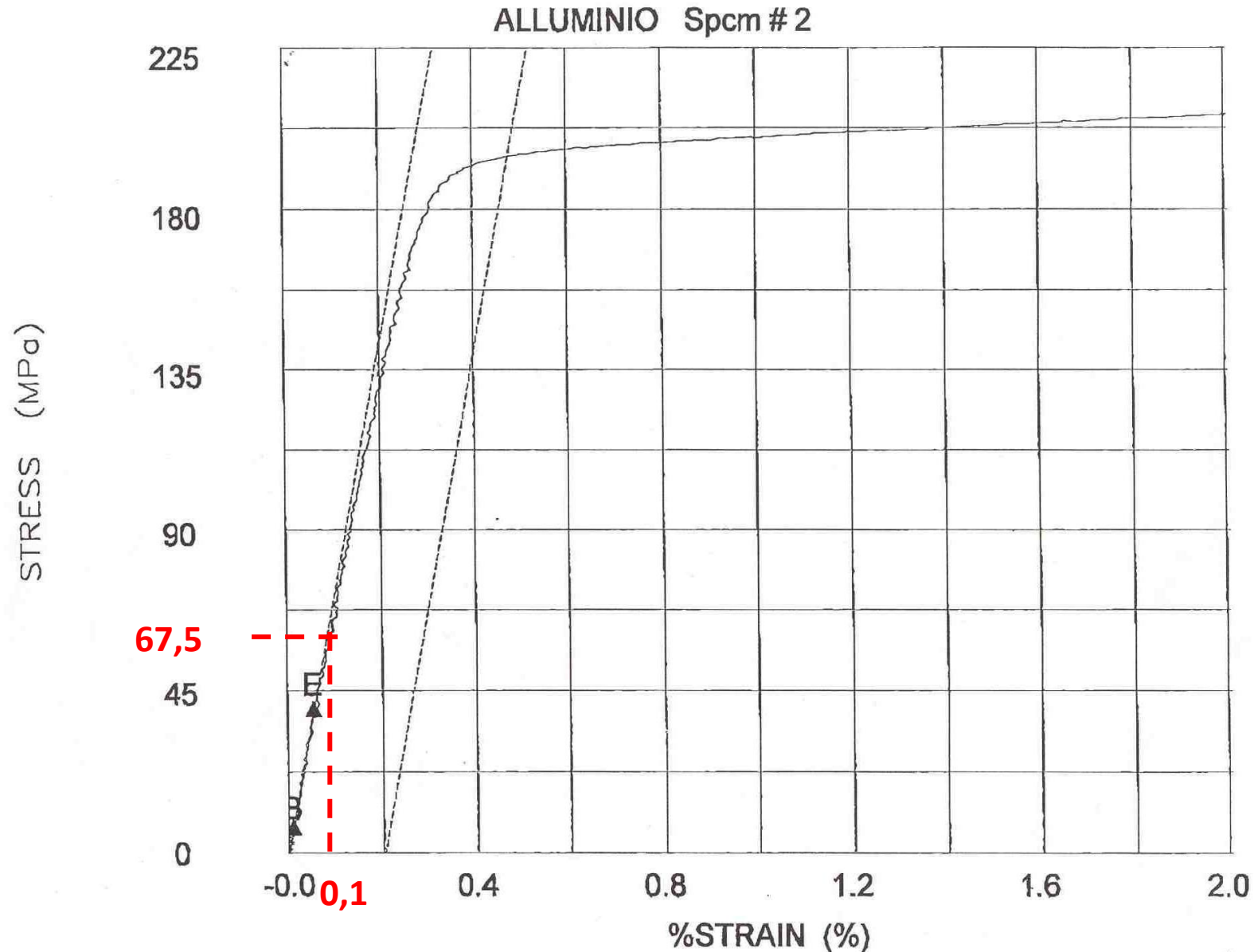
During loading, the total deformation ϵ includes both the elastic and plastic parts. When the load is removed (unloading), the elastic deformation recovered, while the plastic deformation remains this is the permanent deformation of the material.

Exercise 3: Determine the Elastic Modulus of Aluminum and High Density Polyethylene (HDPE) from the graphs below. Comment the obtained results.



Exercise 3 – case 1: Aluminum

Elastic modulus (E) = slope of the linear (elastic) tract of the σ - ε curve



$$\varepsilon\% = 0,1\%$$
$$\rightarrow \varepsilon = 0,001$$

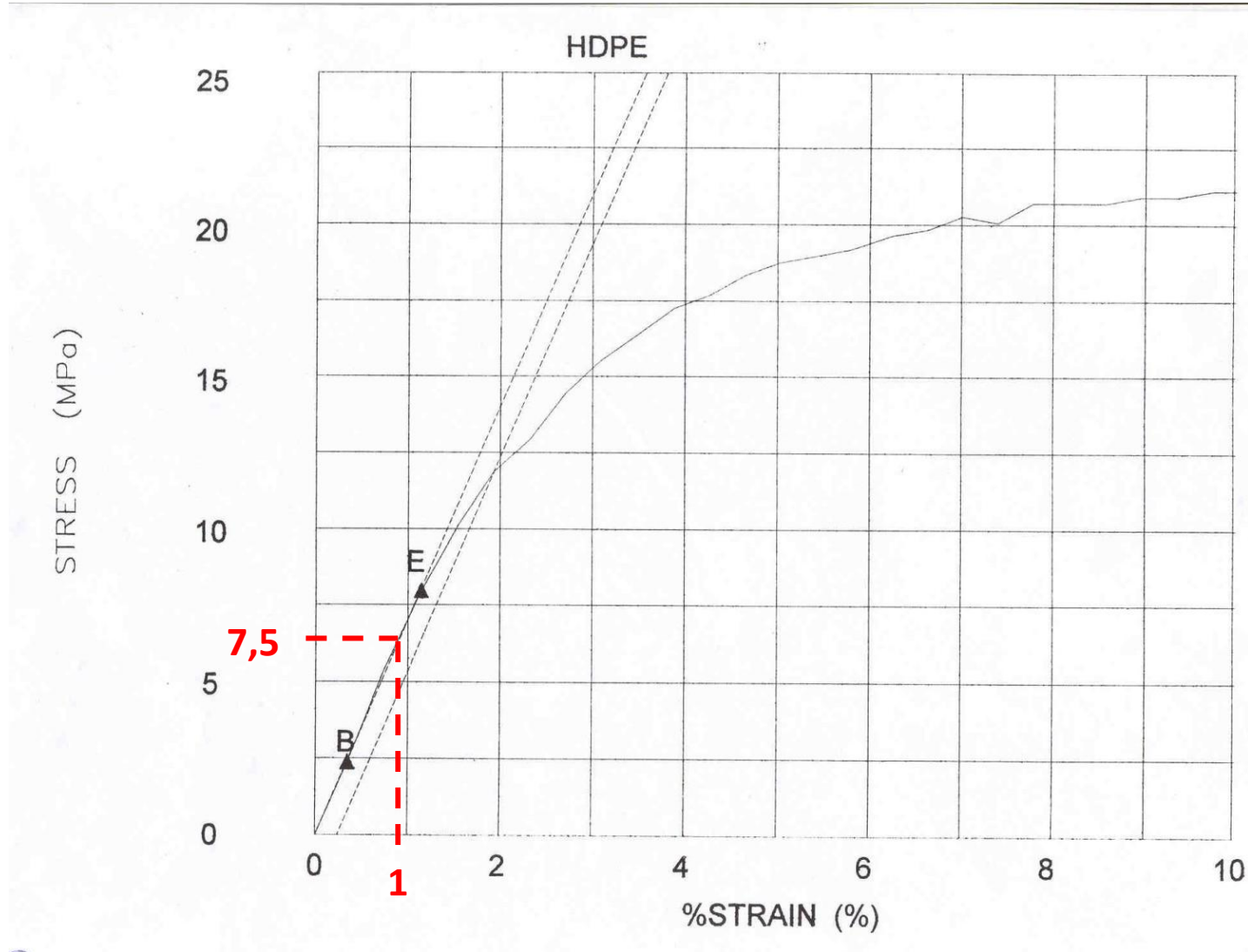
$$E = \sigma / \varepsilon = 67,5$$
$$\text{MPa} / 0,001 = 67500 \text{ MPa}$$

$$1 \text{ GPa} = 1000 \text{ MPa}$$

$$\rightarrow 67500 / 1000 = \mathbf{67,5 \text{ GPa}}$$

Exercise 3 – case 2: HDPE

Elastic modulus (E) = slope of the linear (elastic) tract of the σ - ε curve



$$\varepsilon\% = 1\% \rightarrow \varepsilon = 0,01$$

$$E = \sigma / \varepsilon = 7,5 \text{ MPa} / 0,01 \\ = \textbf{750} \text{ Mpa} \rightarrow 0,75 \text{ GPa}$$

Exercise 3 comparison Aluminum – HDPE

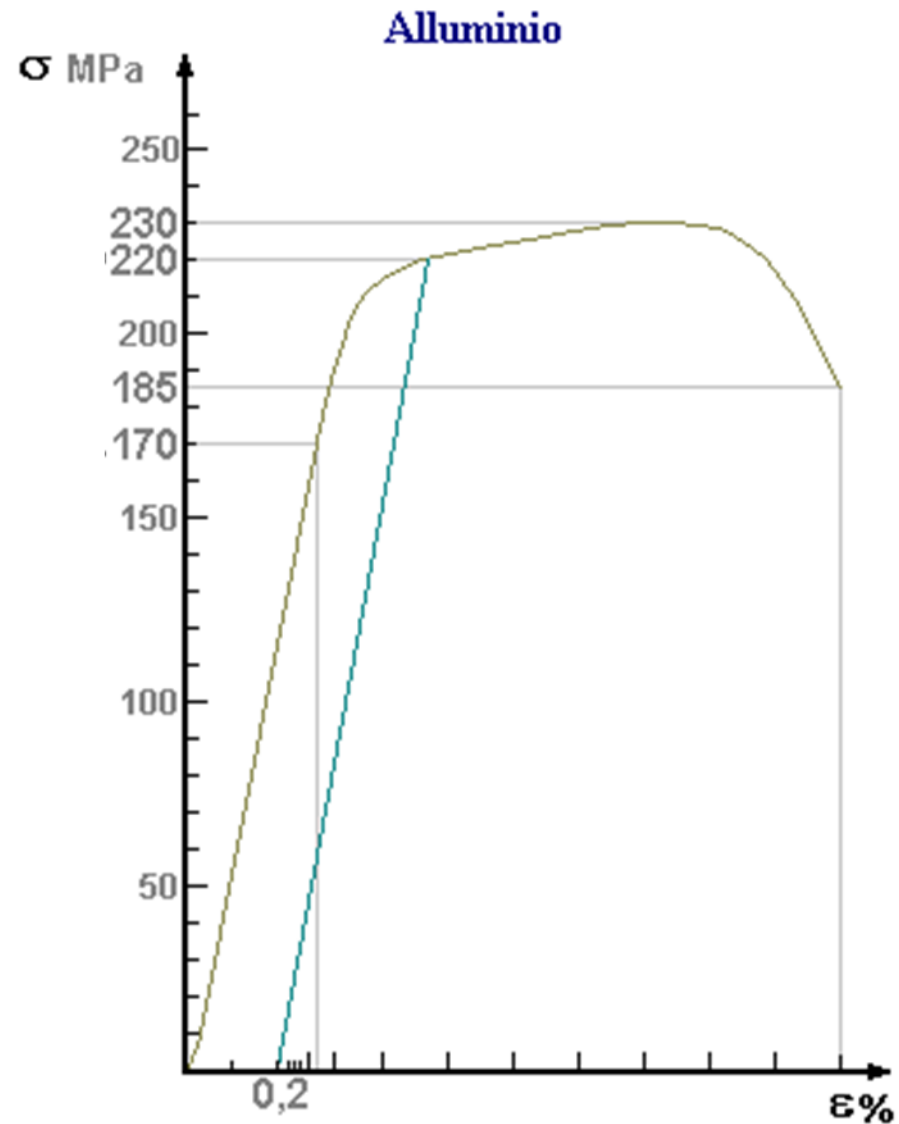
$E_{Al} = 67.5 \text{ GPa}$

$E_{HDPE} = 0.75 \text{ GPa}$

1 order of magnitude difference

Elastic modulus depends on the bonding strength: metallic bond vs van der Waals interactions

Quiz 1

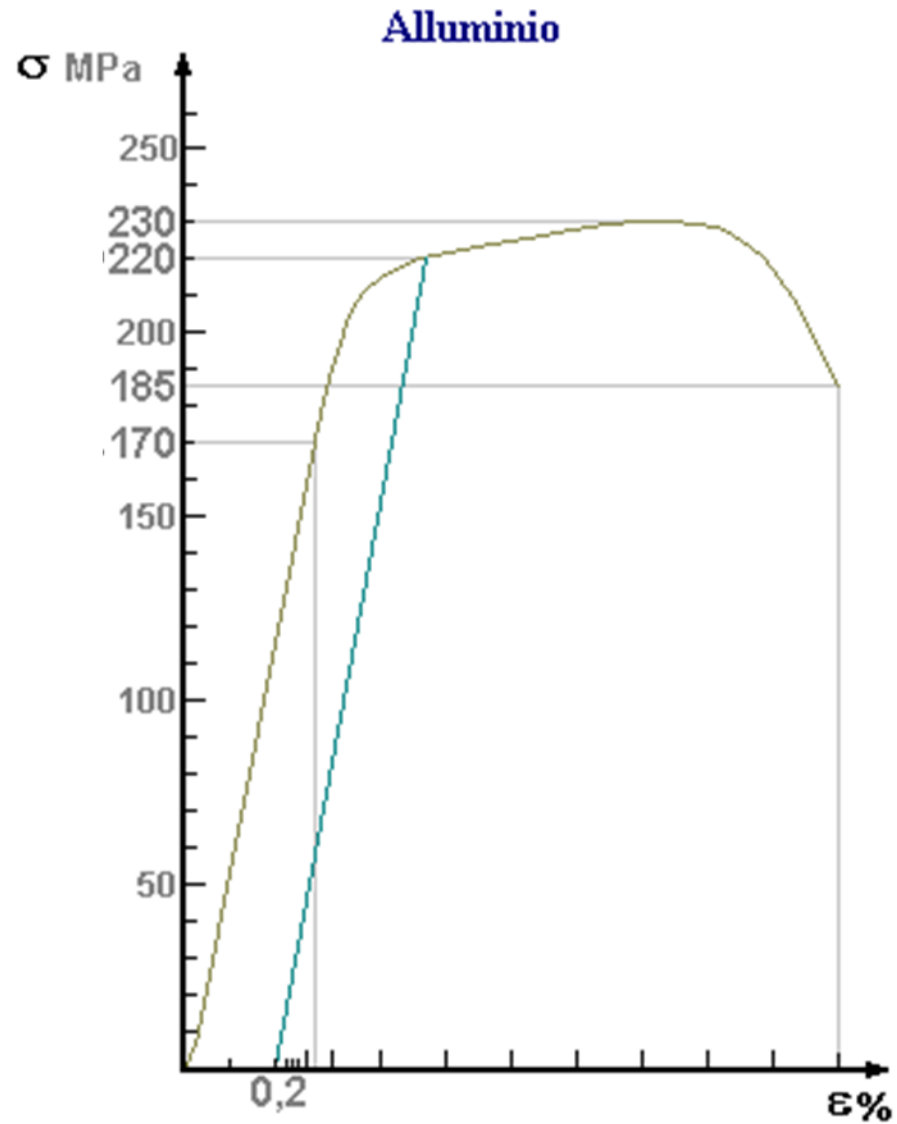


Consider the σ - ϵ curve reported in the figure.

The yield strength (0.2 % offset) is:

- a) 230 Mpa
- b) 220 Mpa
- c) 185 MPa

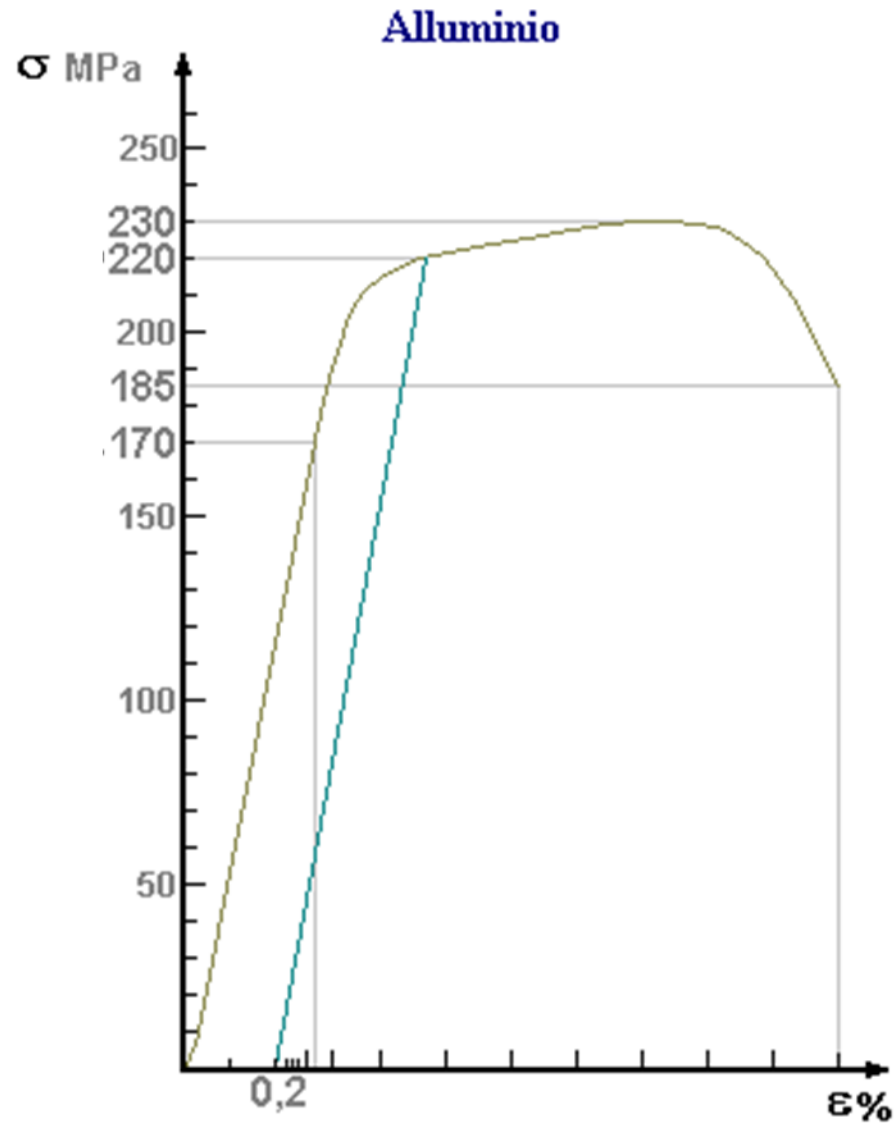
Quiz 1



Consider the σ - ϵ curve reported in the figure. The yield strength (0.2 % offset) is:

- a) 230 Mpa → it's the tensile strength
- b) 220 Mpa
- c) 185 MPa → fracture of the sample

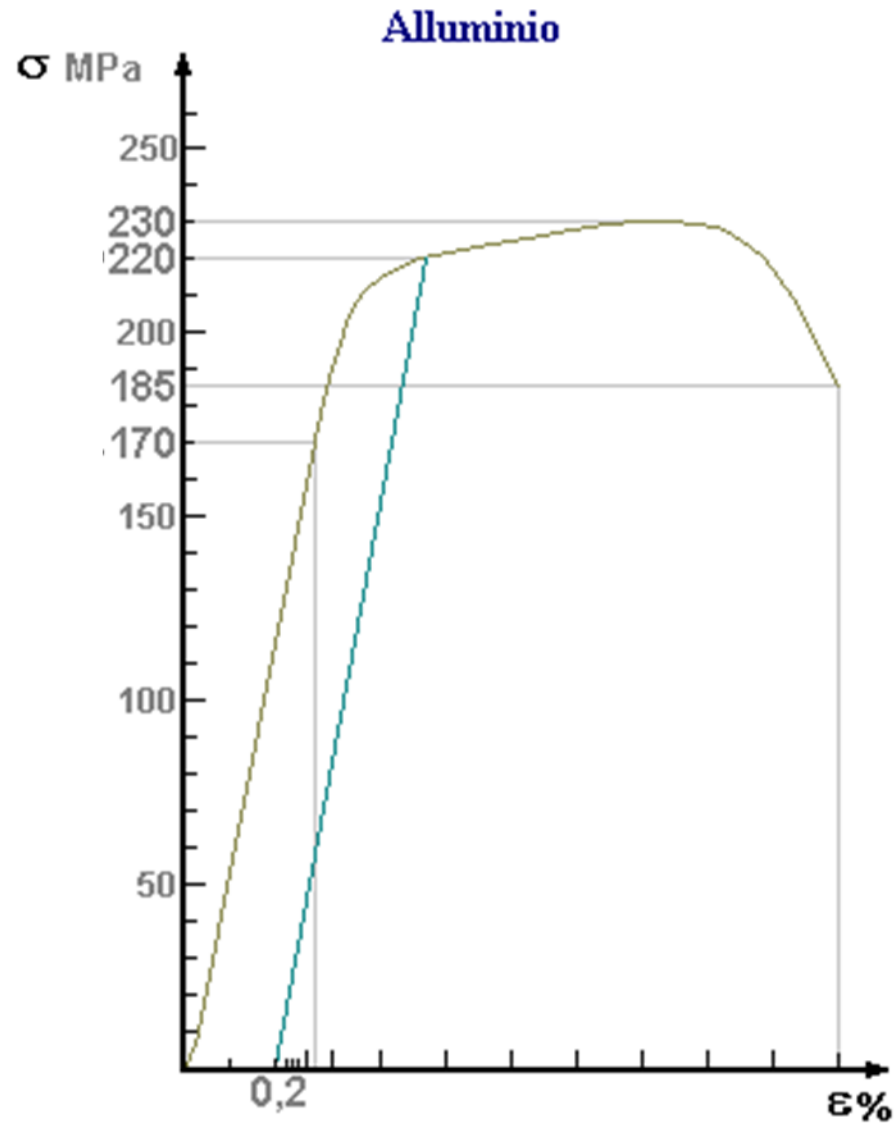
Quiz 2



The σ - ϵ curve in figure comes from the tensile test of an aluminum specimen. Select the right sentence (only one):

- a) The specimen underwent plastic deformation
- b) The specimen underwent fragile fracture
- c) The specimen did not have permanent deformations

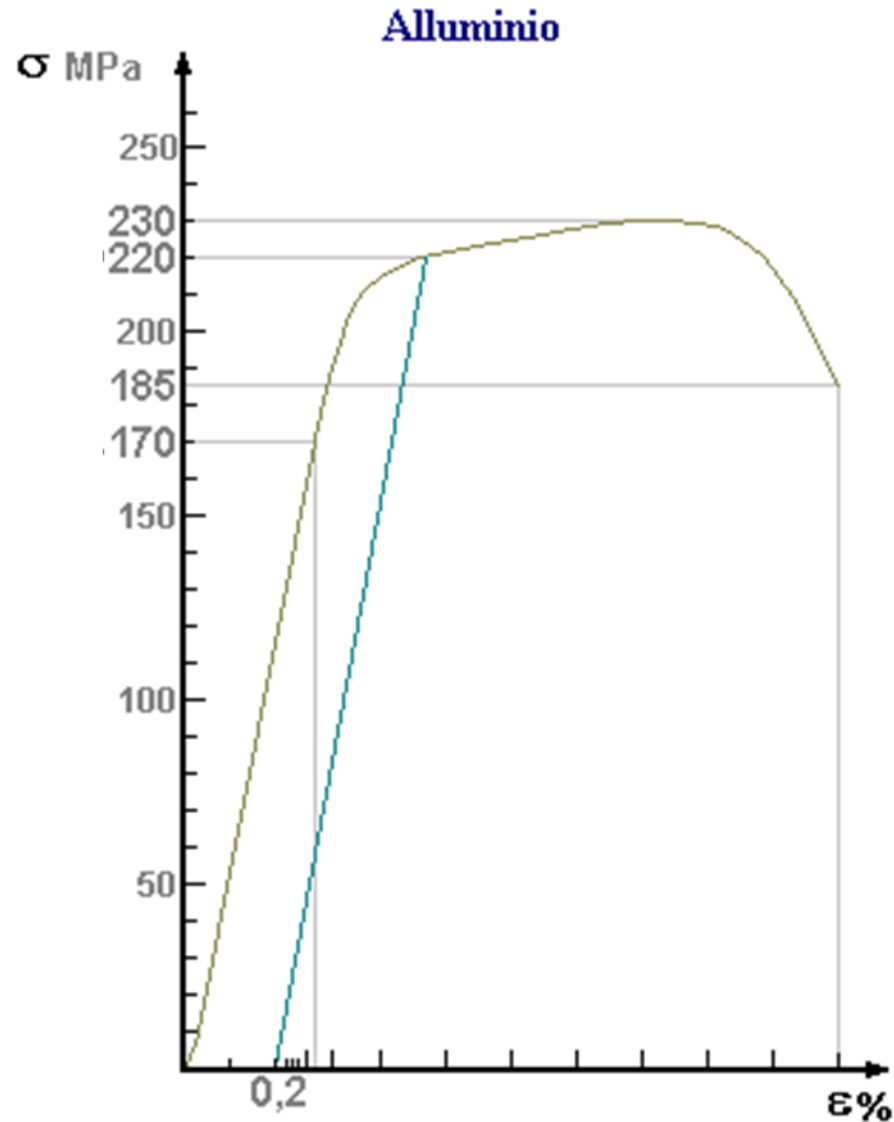
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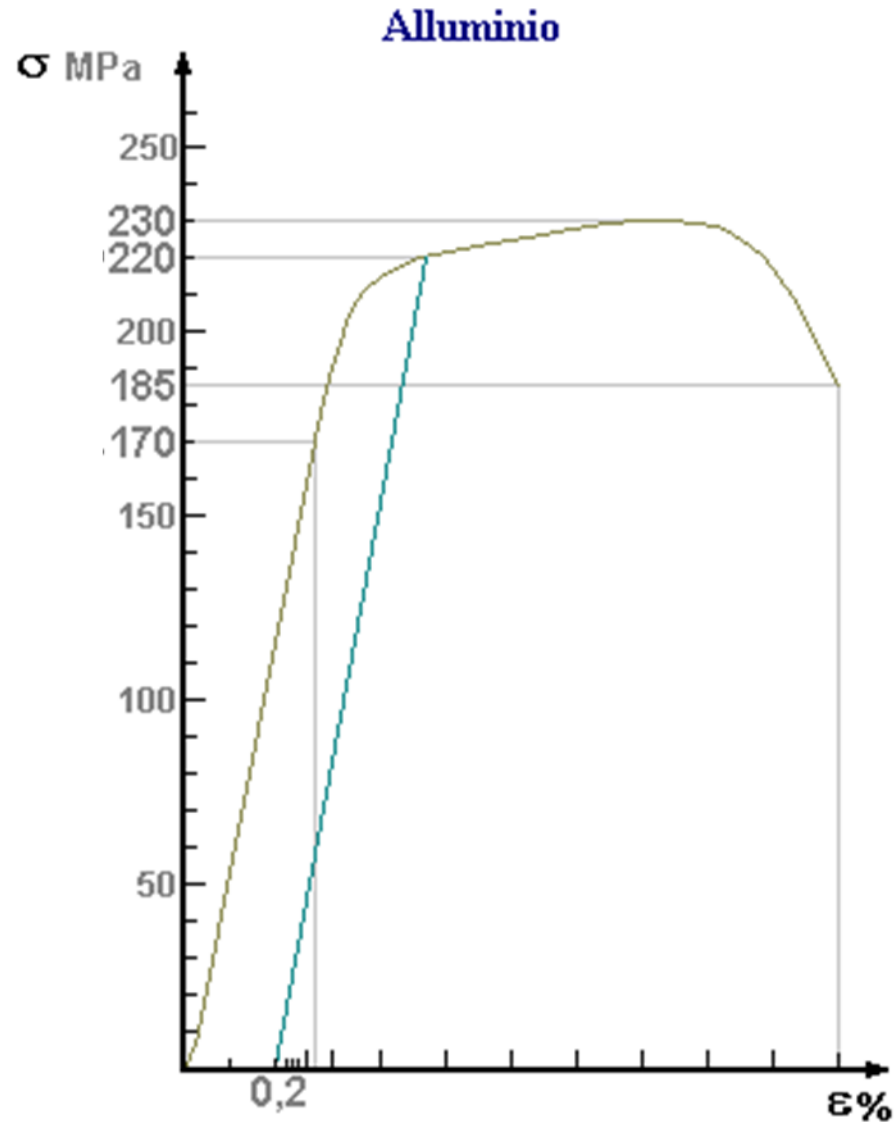
Quiz 3



The σ - ϵ curve in figure comes from the tensile test of an aluminum specimen. Select the right sentence (only one):

- a) The specimen had only elastic deformation
- a) The specimen underwent necking
- a) The yield strenght was not reached in the test

Quiz 3



The σ - ϵ curve in figure comes from the tensile test of an aluminum specimen. Select the right sentence (only one):

- a) The specimen had only elastic deformation
- b) The specimen underwent necking
- c) The yield strenght was not reached in the test